



LUKMAN HAKIM (he/him)

Geospatial & Earth Observation Specialist

M.Sc. Photogrammetry and Geoinformatics | DAAD-EPOS Scholarship Awardee

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PROFILE

Geospatial and Earth Observation professional with 6+ years of experience in GIS, remote sensing, spatial modelling, environmental monitoring, conservation planning, and geospatial data management. I work with satellite and aerial imagery, large spatial datasets, machine-learning workflows, and spatial decision-support methods to translate complex geospatial information into reliable operational and research outputs.

CORE COMPETENCIES

Earth Observation & Remote Sensing

Satellite and aerial imagery, PlanetScope, Sentinel-2, Landsat, multispectral analysis, spectral indices, image classification, change detection, Google Earth Engine, and environmental monitoring.

Geospatial Programming & Data

Python, GeoPandas, Rasterio, GDAL, Pandas, SQL, PostgreSQL/PostGIS, machine learning, workflow automation, reproducible processing, and geospatial data QA/QC.

GIS & Spatial Modelling

QGIS, ArcGIS Pro, spatial suitability analysis, conservation prioritisation, network analysis, raster/vector modelling, spatial statistics, and multi-source geodata integration.

WebGIS & Spatial Data Infrastructure

Leaflet, Flask, GeoServer, REST APIs, OGC services, WebGIS applications, spatial databases, interactive mapping, and automated geospatial services.

PROFESSIONAL EXPERIENCE

(08/2026 – Present)

Stuttgart Technical University of Applied Sciences — Research Associate

in collaboration with the Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL-BW)

- Extending previous research on automated green-roof detection and photovoltaic (PV) mapping, including the identification of existing installations and potential roof areas across Baden-Württemberg, by integrating high-resolution aerial imagery with LoD2 building and roof-plane data under U-Net architecture.

(06/2025 – 11/2025)

Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL-BW) — Student Job

Identification of green roof areas in Baden-Württemberg using LoD2 building data and PlanetScope imagery

- Developed a state-scale Earth Observation and machine-learning workflow combining 14+ million LoD2 roof planes with multispectral PlanetScope imagery, Python, and Random Forest classification to produce a probability-based green-roof inventory for Baden-Württemberg.

(07/2019 – 08/2024)

The Nature Conservancy Indonesia — GIS & Remote Sensing Specialist

- Supported cross-NGO and government collaborations in designing Marine Protected Area networks using MARXAN, gap analysis, and spatial prioritization to inform local, national, and transboundary zoning aligned with Aichi biodiversity conservation targets.
- Developed GIS and remote-sensing workflows for mangrove mapping, including distribution, canopy density, and aboveground biomass estimation to support forest carbon inventory, social forestry initiatives, and REDD+ readiness in Indonesia.
- Developed remote-sensing workflows for benthic habitat classification, including coral, seagrass, algae, sand, rubble, and rock classes, supporting blue carbon initiative
- Applied multi-temporal satellite datasets, including Maxar, SPOT, Landsat, Sentinel, PlanetScope, and Planet NICFI, using Google Earth Engine for environmental monitoring, land-cover analysis, and spatial modelling.

- Planned and conducted aerial data acquisition using small UAS drones and fixed-wing platforms, including flight planning, image triangulation, orthophoto correction, image stitching, and land-use/land-cover classification for coastal community landscapes.
- Integrated heterogeneous geospatial datasets from satellite imagery, field surveys, participatory mapping, administrative boundaries, and open-source geospatial products into structured spatial databases for analysis and decision-making.
- Mentored junior staff in GIS, remote sensing, data management, spatial analysis, and reproducible mapping workflows, improving internal technical capacity across conservation projects.

(08/2018 – 09/2018)

Research Institute for Tuna Fisheries Bali — Internship

- Collected fisheries-related spatial and environmental data for bachelor thesis research.
- Applied remote sensing and Generalized Additive Model (GAM) methods for distribution modeling of Yellowfin Tuna (*Thunnus albacares*) in the Indian Ocean.

EDUCATION

(2024 – 2026)

[Master of Science \(M.Sc.\), Photogrammetry and Geoinformatics — Stuttgart Technical University of Applied Sciences](#)

(DAAD-EPOS Scholarship Awardee • Final Grade 1.8)

(2014 – 2019)

[Bachelor of Science \(B.Sc.\), Cartography and Remote Sensing — Universitas Gadjah Mada](#)

(BIDIKMISI Scholarship Awardee • GPA 3.60)

LANGUAGES

Javanese: Native | Indonesian: Native | English: C1 | German: B1 | Japanese: Novice

SELECTED PUBLICATIONS

- Aung Kyaw, H., Ayala Rico, J., Awontemi, I. A., **et al.** (2026). [Urban Vegetation Mapping from High-Resolution Aerial Imagery with Deep Learning](#). DGPF Jahrestagung 2026.
- Fajariyanto, Y., Green, A., **Hakim, L.**, et al. (2024). [A Resilient Marine Protected Area Network Design: A First for the Arafura–Timor Seas. Coastal Management](#), 52(3), 147–173.
- Prakoso, D. A., **Hakim, L.**, et al. (2023). [The Dynamics of Mangrove and Pond Changes in East Kalimantan, Indonesia. IOP Conference Series: Earth and Environmental Science](#).
- Fajariyanto, Y., Green, A. L., Ramadyan, F., Suardana, N., **Hakim, L.**, et al. (2019). [Designing a Network of Marine Protected Areas for Fisheries Management Area 715 and Six Associated Provinces in Indonesia](#). Technical report prepared for the USAID Sustainable Ecosystems Advanced Project.
- **Hakim, L.**, Lazuardi, W., et al. (2018). [Assessing WorldView-2 Satellite Imagery Accuracy for Bathymetry Mapping in Pahawang Island, Indonesia](#). IOP Conference Series: Earth and Environmental Science.

ADDITIONAL QUALIFICATIONS

Small UAS Remote Pilot • Open Water Diver • Field-based coastal and marine survey experience

EXTRACURRICULAR

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| 2023 | Presenter at 5 th Asia Pacific Coral Reef Symposium, NUS Singapore |
| 2023 | Participant at Mangrove Monitoring International Training (MMIT), Archipelagic Island State Forum Bali Indonesia |
| 2017 | Presenter at World Blue Carbon Conference, Jakarta Convention Center Indonesia |
| 2017 | Presenter at Seminar Nasional Geomatika, Geospatial Information Agency Indonesia |

REFERENCES

- Prof. Dr. rer. nat. Michael Mommert | Hochschule für Technik Stuttgart | michael.mommert@hft-stuttgart.de
- Maryse Hilemann | LGL-BW | Maryse.Hillemann@lgl.bwl.de
- Yusuf Fajariyanto | The Nature Conservancy Indonesia | yfajariyanto@tnc.org